PETTER NILSEN

Cybernetics and robotics engineer

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Ole tilseths veg 2c, 7037 Trondheim Trondheim, Norway
pettnil2001.github.io/petternilsen-portfolio/



EXPERIENCE

Research Assistant
NTNU

Aug 2026 – Present Trondheim

- Support teaching, laboratory sessions and assignments in control systems and guidance, navigation and control.

Systems engineer
PQtech

August 2026 – Present

- Designed PCBs and measurement electronics for monitoring systems.
- Conducted field measurements of power consumption and vibration.
- Processed and analyzed multi-terabyte GPS, IMU and power datasets, including noisy sensor data.

Development Engineer and Master thesis
Equanostic AS

Jul 2024 – Jul 2026 Trondheim

- Developed embedded electronics and PCBs for robotic pipe-inspection systems using ultrasound and radar.
- Developed an inertial navigation solution combining IMU, encoder and computer-vision measurements.

Bachelor Thesis
ScoutDI

Jan 2024 – May 2024 Trondheim

- Production development of ultrasonic thickness measurement systems for drones. The developed product got certified patented.

Drone Developer
Ascend NTNU

Sep 2022 – Jun 2023 Trondheim

- Designed, integrated and tested hardware modules for autonomous and manually operated drones.

ADDITIONAL EXPERIENCE

Student Assistant, NTNU Aug 2022 – Jun 2025
Firefighter, Norwegian Air Force Oct 2020 – Jul 2021

PROFILE

Cybernetics and robotics engineer with experience in state estimation, sensor fusion and embedded development for autonomous and robotic systems. Worked with IMU, encoders and computer vision in state estimation and navigation, in addition to large-scale sensor-data analysis.

CERTIFICATIONS

Driving License BE
Car and heavy trailer

Fire and Safety
Certified firefighter training

TECHNICAL SKILLS

Hard-working | Eye for detail | Creative

Robotics | State estimation | SLAM | PCB Design | Motor Control

ROS2 | Fusion 360 | Altium Designer | Python | C++

LANGUAGES

Norwegian – Native
English – Fluent
German – Basic

EDUCATION

M.Sc. Cybernetics and Robotics
NTNU

Aug 2024 – Jun 2026 Trondheim

B.Sc. Electrical Engineering, Automation and Robotics
NTNU

Aug 2021 – Jun 2024 Trondheim